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1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>StudyMate - AI-Powered PDF Q&A System</title>
7   <script src="https://cdn.tailwindcss.com"></script>
8   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/tailwindcss@2.2.19/dist/tailwind.min.css">
9   <style>
10     @import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap');
11
12     * {
13       font-family: 'Inter', sans-serif;
14     }
15
16     .gradient-bg {
17       background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
18     }
19
20     .card-gradient {
21       background: linear-gradient(145deg, #ffffff 0%, #f8f9fa 100%);
22       box-shadow: 0 10px 25px rgba(0, 0, 0, 0.1);
23     }
24
25     .upload-area {
26       border: 2px dashed #cbd5e0;
27       transition: all 0.3s ease;
28     }
29
30     .upload-area:hover {
31       border-color: #667eea;
32       background-color: #f7fafc;
33     }
34
35     .chat-bubble {
36       max-width: 80%;
37       border-radius: 18px;
38       padding: 12px 16px;
39       margin: 8px 0;
40     }
41
42     .user-bubble {
43       background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
44       color: white;
45       margin-left: auto;
46     }
47
48     .ai-bubble {
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49         background: #f1f5f9;
50         color: #374151;
51     }
52
53     .typing-indicator {
54         display: inline-flex;
55         align-items: center;
56     }
57
58     .typing-dot {
59         width: 6px;
60         height: 6px;
61         background-color: #9ca3af;
62         border-radius: 50%;
63         margin: 0 2px;
64         animation: typing 1.4s infinite ease-in-out;
65     }
66
67     .typing-dot:nth-child(2) {
68         animation-delay: 0.2s;
69     }
70
71     .typing-dot:nth-child(3) {
72         animation-delay: 0.4s;
73     }
74
75     @keyframes typing {
76         0%, 60%, 100% {
77             transform: translateY(0);
78         }
79         30% {
80             transform: translateY(-5px);
81         }
82     }
83
84     .fade-in {
85         animation: fadeIn 0.5s ease-in;
86     }
87
88     @keyframes fadeIn {
89         from { opacity: 0; transform: translateY(10px); }
90         to { opacity: 1; transform: translateY(0); }
91     }
92 </style>
93 </head>
94 <body class="min-h-screen bg-gray-50">
95     <!-- Header -->
96     <header class="gradient-bg text-white shadow-lg">
97         <div class="container mx-auto px-4 py-6">
98             <div class="flex items-center justify-between">
99                 <div class="flex items-center space-x-3">

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100         <div class="w-10 h-10 bg-white rounded-lg flex items-center
justify-center">
101             <i class="fas fa-graduation-cap text-purple-600 text-
xl"></i>
102         </div>
103         <h1 class="text-2xl font-bold">StudyMate</h1>
104     </div>
105     <nav class="hidden md:flex space-x-6">
106         <a href="#" class="hover:text-purple-200
transition">Features</a>
107         <a href="#" class="hover:text-purple-200 transition">How It
Works</a>
108         <a href="#" class="hover:text-purple-200
transition">Pricing</a>
109     </nav>
110     <button class="bg-white text-purple-600 px-4 py-2 rounded-lg
font-semibold hover:bg-purple-50 transition">
111         Get Started
112     </button>
113 </div>
114 </div>
115 </header>
116
117 <!-- Main Content -->
118 <main class="container mx-auto px-4 py-8">
119     <!-- Hero Section -->
120     <section class="text-center mb-12">
121         <h2 class="text-4xl md:text-5xl font-bold text-gray-800 mb-4">
AI-Powered Study Assistant
122         </h2>
123         <p class="text-xl text-gray-600 max-w-3xl mx-auto mb-8">
Upload your PDF textbooks and research papers, then ask
124 questions to get instant, intelligent answers powered by artificial
125 intelligence.
126         </p>
127
128     <!-- Upload Area -->
129     <div class="max-w-2xl mx-auto mb-12">
130         <div class="card-gradient rounded-2xl p-8 upload-area"
id="uploadArea">
131             <div class="text-center">
132                 <div class="w-16 h-16 bg-purple-100 rounded-full flex
items-center justify-center mx-auto mb-4">
133                     <i class="fas fa-file-pdf text-purple-600 text-
2xl"></i>
134                 </div>
135                 <h3 class="text-xl font-semibold text-gray-800 mb-
2">Upload Your PDF</h3>
136                 <p class="text-gray-600 mb-4">Drag and drop your PDF
file here, or click to browse</p>

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137         <input type="file" id="pdfUpload" accept=".pdf"
class="hidden">
138         <label for="pdfUpload" class="bg-purple-600 text-white
px-6 py-3 rounded-lg font-semibold cursor-pointer hover:bg-purple-700
transition inline-block">
139             Choose File
140         </label>
141         <p class="text-sm text-gray-500 mt-2">Max file size:
10MB</p>
142     </div>
143 </div>
144 </div>
145 </section>
146
147 <!-- Chat Interface -->
148 <section class="max-w-4xl mx-auto">
149     <div class="card-gradient rounded-2xl overflow-hidden shadow-xl">
150         <div class="bg-gradient-to-r from-purple-600 to-purple-800
text-white p-6">
151             <h3 class="text-xl font-semibold">Ask Questions About Your
Document</h3>
152             <p class="text-purple-200">The AI assistant is ready to
help you understand your material</p>
153         </div>
154
155         <div class="p-6">
156             <!-- Chat Messages -->
157             <div id="chatContainer" class="h-96 overflow-y-auto mb-4
space-y-4 p-4 bg-gray-50 rounded-lg">
158                 <div class="ai-bubble chat-bubble fade-in">
159                     <div class="flex items-start space-x-2">
160                         <div class="w-8 h-8 bg-purple-600 rounded-full
flex items-center justify-center flex-shrink-0">
161                             <i class="fas fa-robot text-white text-sm">
</i>
162                         </div>
163                         <div>
164                             <p class="font-medium text-gray-
700">StudyMate AI</p>
165                             <p class="text-gray-600">Hello! I'm your AI
study assistant. Upload a PDF document and I'll help you understand its content
through interactive Q&A.</p>
166                         </div>
167                     </div>
168                 </div>
169
170                 <!-- Sample questions -->
171                 <div class="text-center text-sm text-gray-500 my-4">
172                     <p>Try asking:</p>
173                     <div class="flex flex-wrap justify-center gap-2 mt-
2">

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174         <button onclick="askQuestion('What is the main
topic of this document?')" class="bg-gray-100 text-gray-700 px-3 py-1 rounded-
full text-xs hover:bg-gray-200 transition">
175             What is the main topic?
176         </button>
177         <button onclick="askQuestion('Summarize the key
points')" class="bg-gray-100 text-gray-700 px-3 py-1 rounded-full text-xs
hover:bg-gray-200 transition">
178             Summarize key points
179         </button>
180         <button onclick="askQuestion('Explain the most
important concept')" class="bg-gray-100 text-gray-700 px-3 py-1 rounded-full
text-xs hover:bg-gray-200 transition">
181             Explain important concept
182         </button>
183     </div>
184 </div>
185 </div>
186
187     <!-- Input Area -->
188     <div class="flex space-x-3">
189         <input
190             type="text"
191             id="questionInput"
192             placeholder="Ask a question about your document ..."
193             class="flex-1 border border-gray-300 rounded-lg px-
4 py-3 focus:outline-none focus:ring-2 focus:ring-purple-500 focus:border-
transparent"
194             disabled
195         >
196         <button
197             id="askButton"
198             class="bg-purple-600 text-white px-6 py-3 rounded-
lg font-semibold hover:bg-purple-700 transition disabled:opacity-50
disabled:cursor-not-allowed"
199             disabled
200         >
201             <i class="fas fa-paper-plane mr-2"></i>Ask
202         </button>
203     </div>
204 </div>
205 </div>
206 </section>
207
208     <!-- Features Section -->
209     <section class="mt-16">
210         <h2 class="text-3xl font-bold text-center text-gray-800 mb-
12">Powerful Features for Students</h2>
211         <div class="grid md:grid-cols-3 gap-8">
212             <div class="card-gradient rounded-xl p-6 text-center">

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213         <div class="w-12 h-12 bg-purple-100 rounded-full flex
items-center justify-center mx-auto mb-4">
214             <i class="fas fa-brain text-purple-600 text-xl"></i>
215         </div>
216         <h3 class="text-xl font-semibold mb-3">AI-Powered
Understanding</h3>
217         <p class="text-gray-600">Advanced AI analyzes your
documents and provides accurate, contextual answers to your questions.</p>
218         </div>
219
220         <div class="card-gradient rounded-xl p-6 text-center">
221             <div class="w-12 h-12 bg-purple-100 rounded-full flex
items-center justify-center mx-auto mb-4">
222                 <i class="fas fa-search text-purple-600 text-xl"></i>
223             </div>
224             <h3 class="text-xl font-semibold mb-3">Smart Document
Analysis</h3>
225             <p class="text-gray-600">Extract key concepts, summaries,
and important information from complex academic materials.</p>
226             </div>
227
228             <div class="card-gradient rounded-xl p-6 text-center">
229                 <div class="w-12 h-12 bg-purple-100 rounded-full flex
items-center justify-center mx-auto mb-4">
230                     <i class="fas fa-comments text-purple-600 text-xl"></i>
231                 </div>
232                 <h3 class="text-xl font-semibold mb-3">Interactive
Learning</h3>
233                 <p class="text-gray-600">Engage in natural conversations
about your study materials to deepen your understanding.</p>
234                 </div>
235             </div>
236         </section>
237     </main>
238
239     <!-- Footer -->
240     <footer class="bg-gray-800 text-white mt-16 py-12">
241         <div class="container mx-auto px-4">
242             <div class="grid md:grid-cols-4 gap-8">
243                 <div>
244                     <h4 class="text-lg font-semibold mb-4">StudyMate</h4>
245                     <p class="text-gray-400">Your AI-powered study companion
for smarter learning and better understanding.</p>
246                 </div>
247
248                 <div>
249                     <h4 class="text-lg font-semibold mb-4">Features</h4>
250                     <ul class="space-y-2 text-gray-400">
251                         <li><a href="#" class="hover:text-white transition">PDF
Analysis</a></li>

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252         <li><a href="#" class="hover:text-white transition">Q&A
System</a></li>
253         <li><a href="#" class="hover:text-white
transition">Study Assistance</a></li>
254     </ul>
255 </div>
256
257     <div>
258         <h4 class="text-lg font-semibold mb-4">Resources</h4>
259         <ul class="space-y-2 text-gray-400">
260             <li><a href="#" class="hover:text-white
transition">Documentation</a></li>
261             <li><a href="#" class="hover:text-white
transition">Tutorials</a></li>
262             <li><a href="#" class="hover:text-white
transition">API</a></li>
263         </ul>
264     </div>
265
266     <div>
267         <h4 class="text-lg font-semibold mb-4">Connect</h4>
268         <div class="flex space-x-4">
269             <a href="#" class="w-10 h-10 bg-gray-700 rounded-full
flex items-center justify-center hover:bg-purple-600 transition">
270                 <i class="fab fa-twitter"></i>
271             </a>
272             <a href="#" class="w-10 h-10 bg-gray-700 rounded-full
flex items-center justify-center hover:bg-purple-600 transition">
273                 <i class="fab fa-linkedin"></i>
274             </a>
275             <a href="#" class="w-10 h-10 bg-gray-700 rounded-full
flex items-center justify-center hover:bg-purple-600 transition">
276                 <i class="fab fa-github"></i>
277             </a>
278         </div>
279     </div>
280 </div>
281
282 <div class="border-t border-gray-700 mt-8 pt-8 text-center text-
gray-400">
283     <p>&copy; 2024 StudyMate. All rights reserved.</p>
284 </div>
285 </div>
286 </footer>
287
288 <script>
289     // DOM Elements
290     const pdfUpload = document.getElementById('pdfUpload');
291     const uploadArea = document.getElementById('uploadArea');
292     const questionInput = document.getElementById('questionInput');
293     const askButton = document.getElementById('askButton');

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294     const chatContainer = document.getElementById('chatContainer');
295
296     // Sample responses for demonstration
297     const sampleResponses = {
298         'What is the main topic of this document?': 'Based on my analysis,
this document appears to be about machine learning fundamentals, covering
topics such as supervised learning, neural networks, and data preprocessing
techniques. The main focus is on providing a comprehensive introduction to AI
concepts for beginners.',
299         'Summarize the key points': 'The key points in this document
include: 1) Introduction to artificial intelligence and its applications, 2)
Detailed explanation of machine learning algorithms, 3) Discussion on neural
network architectures, 4) Data preprocessing and feature engineering
techniques, and 5) Ethical considerations in AI development.',
300         'Explain the most important concept': 'The most important concept
discussed is "Supervised Learning." This involves training algorithms on
labeled datasets where both input data and corresponding outputs are provided.
The model learns to map inputs to outputs and can then make predictions on new,
unseen data. This forms the foundation for many practical AI applications.',
301         'What are the applications mentioned?': 'The document mentions
several applications including image recognition systems, natural language
processing for chatbots, predictive analytics in healthcare, autonomous
vehicles, and recommendation systems used by streaming services and e-commerce
platforms.',
302         'How does neural network work?': 'Neural networks work by
simulating the human brain\'s neural connections. They consist of layers of
interconnected nodes (neurons) that process information. Each connection has a
weight that adjusts during training, allowing the network to learn complex
patterns and make accurate predictions through forward and backward propagation
processes.'
303     };
304
305     // Default response for unknown questions
306     const defaultResponse = 'I\'ve analyzed your document and found it
contains technical academic content. Could you please clarify which specific
aspect you\'d like me to focus on? I can provide detailed explanations about
concepts, summaries, or help with understanding complex topics.';
307
308     // Handle file upload
309     pdfUpload.addEventListener('change', function(e) {
310         const file = e.target.files[0];
311         if (file && file.type === 'application/pdf') {
312             // Simulate file processing
313             uploadArea.innerHTML = `
314                 <div class="text-center">
315                     <div class="w-16 h-16 bg-green-100 rounded-full flex
items-center justify-center mx-auto mb-4">
316                         <i class="fas fa-check text-green-600 text-2xl">
</i>
317                     </div>

```

```

318         <h3 class="text-xl font-semibold text-gray-800 mb-
2">Document Uploaded Successfully!</h3>
319         <p class="text-gray-600 mb-2">${file.name}</p>
320         <p class="text-sm text-green-600">Ready for
analysis</p>
321     </div>
322     `;
323
324     // Enable chat functionality
325     questionInput.disabled = false;
326     askButton.disabled = false;
327
328     // Add success message to chat
329     addMessage('ai', 'Great! I\'ve successfully analyzed your PDF
document. I\'m now ready to answer your questions about its content. What would
you like to know?');
330     } else {
331         alert('Please upload a valid PDF file.');
```

```

365         setTimeout(() => {
366             removeThinkingMessage(thinkingMessage);
367             const response = sampleResponses[question] || defaultResponse;
368             addMessage('ai', response);
369         }, 1500);
370     }
371
372     // Add a message to the chat
373     function addMessage(sender, text) {
374         const messageDiv = document.createElement('div');
375         messageDiv.className = `${sender === 'user' ? 'user-bubble' : 'ai-
bubble'} chat-bubble fade-in`;
376
377         messageDiv.innerHTML = `
378             <div class="flex items-start space-x-2">
379                 <div class="w-8 h-8 ${sender === 'user' ? 'bg-white' : 'bg-
purple-600'} rounded-full flex items-center justify-center flex-shrink-0">
380                     <i class="${sender === 'user' ? 'fas fa-user text-
purple-600' : 'fas fa-robot text-white'} text-sm"></i>
381                 </div>
382                 <div>
383                     <p class="font-medium ${sender === 'user' ? 'text-
white' : 'text-gray-700'}">${sender === 'user' ? 'You' : 'StudyMate AI'}</p>
384                     <p class="${sender === 'user' ? 'text-white' : 'text-
gray-600'}">${text}</p>
385                 </div>
386             </div>
387         `;
388
389         chatContainer.appendChild(messageDiv);
390         chatContainer.scrollTop = chatContainer.scrollHeight;
391         return messageDiv;
392     }
393
394     // Add thinking indicator
395     function addThinkingMessage() {
396         const thinkingDiv = document.createElement('div');
397         thinkingDiv.className = 'ai-bubble chat-bubble';
398         thinkingDiv.id = 'thinking-message';
399
400         thinkingDiv.innerHTML = `
401             <div class="flex items-start space-x-2">
402                 <div class="w-8 h-8 bg-purple-600 rounded-full flex items-
center justify-center flex-shrink-0">
403                     <i class="fas fa-robot text-white text-sm"></i>
404                 </div>
405                 <div>
406                     <p class="font-medium text-gray-700">StudyMate AI</p>
407                     <div class="typing-indicator">
408                         <span class="typing-dot"></span>
409                         <span class="typing-dot"></span>

```

```
410         <span class="typing-dot"></span>
411     </div>
412 </div>
413 </div>
414 `;
415
416     chatContainer.appendChild(thinkingDiv);
417     chatContainer.scrollTop = chatContainer.scrollHeight;
418     return thinkingDiv;
419 }
420
421 // Remove thinking indicator
422 function removeThinkingMessage(thinkingElement) {
423     if (thinkingElement && thinkingElement.parentNode) {
424         thinkingElement.parentNode.removeChild(thinkingElement);
425     }
426 }
427
428 // Drag and drop functionality
429 uploadArea.addEventListener('dragover', function(e) {
430     e.preventDefault();
431     this.style.borderColor = '#667eea';
432     this.style.backgroundColor = '#f7fafc';
433 });
434
435 uploadArea.addEventListener('dragleave', function(e) {
436     e.preventDefault();
437     this.style.borderColor = '#cbd5e0';
438     this.style.backgroundColor = '';
439 });
440
441 uploadArea.addEventListener('drop', function(e) {
442     e.preventDefault();
443     this.style.borderColor = '#cbd5e0';
444     this.style.backgroundColor = '';
445
446     const files = e.dataTransfer.files;
447     if (files.length > 0) {
448         pdfUpload.files = files;
449         const event = new Event('change');
450         pdfUpload.dispatchEvent(event);
451     }
452 });
453
454 // Click anywhere on upload area to trigger file input
455 uploadArea.addEventListener('click', function() {
456     pdfUpload.click();
457 });
458 </script>
459 </body>
460 </html>
```